Task 3: E-Commerce Data Analysis Report

Organization: WeIntern Pvt Ltd
Project Title: E-Commerce Sales and Customer Analysis
Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook

1. Title and Objective

The objective of this project is to clean and analyze e-commerce transaction data to identify sales trends, customer purchasing patterns, and business performance. The analysis uses data cleaning, KPI calculations, and visualizations to generate meaningful business insights.

2. Dataset Overview

The project uses an e-commerce dataset containing transaction and sales information. The dataset is examined to understand its structure, columns, data types, missing values, and duplicate records before performing analysis.

3. Cleaning Summary

The dataset is checked for missing values, duplicate records, invalid quantities, inconsistent categories, and incorrect data formats. Appropriate cleaning methods are applied, and major cleaning decisions are documented to improve data quality and reliability.

4. KPI Summary

The sales dashboard reports the following key performance indicators:

- **Total Revenue:** ₹10,53,736
- **Total Orders:** 287
- **Units Sold:** 857
- **Average Order Value:** ₹3,672

These KPIs provide an overview of sales performance and transaction activity.

5. Trend Analysis

The monthly sales trend shows variations in revenue throughout the observed period. According to the dashboard, July 2026 recorded the highest monthly revenue, while August 2026 showed a noticeable decline. These changes can help identify periods that require further business investigation.

6. Customer Insights

Customer analysis can be used to identify repeat purchases, high-value customers, and differences in purchasing behavior. Customer-level transaction data should be grouped and analyzed to determine which customers contribute most to overall revenue.

7. Visualization Section

The dashboard includes a monthly sales trend line chart, a category-wise revenue bar chart, a city-wise revenue bar chart, and a payment-mode doughnut chart. These visualizations help compare sales performance across time periods, product categories, locations, and payment methods.

Each chart should be accompanied by a short explanation of its main observation and business relevance.

8. Key Findings

The dashboard indicates that Sports is the highest-revenue category and Hyderabad is the leading city by revenue. July 2026 recorded the strongest monthly sales performance. The payment-mode chart also shows the distribution of transactions across several payment methods.

These findings should be verified against the underlying dataset before final submission.

9. Recommendations

- Investigate the reasons for monthly revenue fluctuations.
- Review inventory and promotional strategies for high-performing categories.
- Study city-level sales differences to identify business opportunities.
- Analyze customer purchase frequency and spending patterns.

- Monitor payment preferences to improve the purchasing experience.
- Maintain regular data-quality checks to support reliable reporting.

10. Limitations or Assumptions

The analysis depends on the accuracy and completeness of the available dataset. Missing values, duplicate records, invalid entries, and inconsistent categories may affect the results if they are not handled appropriately. The analysis describes observed data and does not establish the causes of sales changes. All KPI values and findings should be validated against the final cleaned dataset before submission.

Conclusion

This project demonstrates the application of data cleaning, exploratory data analysis, KPI calculation, and data visualization to e-commerce data. The results provide an overview of sales performance and help identify areas for further investigation and business improvement.